Fatemehsadat Mousavinasab, Ph.D.

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LinkedIn | Portfolio | Google Scholar
U.S. Permanent Resident (no sponsorship required) | Spanish (beginner), English (fluent), Persian (native)

SUMMARY

Ph.D. immunologist (graduating Dec 2026 – Mar 2027, GPA 4.19) with 10 publications and hands-on experience across a broad range of laboratory techniques: creating disease models, developing assays, and generating antigen-specific T-cell tools. My research has focused on mucosal immunity, specifically regulatory T cells in autoimmune (celiac disease) and allergic (eosinophilic esophagitis) conditions. I also prepared the quality documentation for a laboratory pursuing ISO 17025 accreditation and supervised an immunology laboratory section, instructing students in both theory and practice of core techniques and mentoring them at the bench. I design and run experiments end to end, establish new assays, and bring scientific problem-solving and a quality mindset to research and development in academic, biotech, and pharmaceutical laboratories.

EDUCATION

Ph.D., Biomedical Sciences, Georgia State University, Atlanta, GA *Expected Dec 2026 – Mar 2027*
GPA 4.19, Dissertation: Peripheral regulatory T cells in oral tolerance to gluten in celiac disease mouse model. Mentor: Leszek Ignatowicz, Ph.D.

M.Sc., Medical Immunology, Shahid Beheshti University of Medical Sciences, Tehran, Iran *2013 – 2016*
GPA 3.86. Thesis: Regulatory T cells in esophageal tissue of patients with eosinophilic esophagitis vs. GERD and controls.

B.Sc., Medical Laboratory Sciences, Arak University of Medical Sciences, Iran *2008 – 2012*
GPA 3.93

RESEARCH AND MENTORSHIP EXPERIENCE

Ph.D. Candidate, Institute for Biomedical Science, Georgia State University *2022 – Present*

- Developed a novel humanized mouse model to test the role of peripheral regulatory T cells in oral tolerance to gluten in celiac disease.
- Designed and ran in vivo studies across different mouse groups.
- Profiled gut T-cell populations (Activated $CD4^+$ cells, Tregs, Th17/Treg balance, cytokines) by multicolor flow cytometry (more than 10 colors), histology, and functional assays.
- Established hybridomas specific for human class II MHC/peptide complexes (thymoma fusion, HLA-DQ8 dendritic-cell co-culture, peptide testing).
- Published step-by-step protocol (Bio-protocol, 2026) so other laboratories can reproduce the hybridoma method.
- Managed mouse colonies; performed injections, gavage, blood sampling, surgery, and organ isolation.
- Trained new students in core laboratory techniques.

Head of Immunology Section & Training Lead, GenIran Research Lab, Tehran, Iran *2020 – 2022*

- Led the immunology section and immunoassay-based research projects.
- Designed and delivered a hands-on immunoassay training course for students and interns, including syllabus development and practical sessions.
- Mentored students in immunoassay workflows, from sample preparation to data interpretation.
- Mentored students in Molecular genetic workflows (PCR, RT-qPCR), from sample preparation to data interpretation.
- Developed SOPs and quality records in line with ISO/IEC 17025 requirements, in preparation for laboratory accreditation

Researcher, Research Institute for Gastroenterology & Liver Diseases, Tehran, Iran *May – Dec 2019*

- Designed and wrote a full research proposal investigating the effect of tofacitinib on monocyte polarization toward M1/M2 macrophages in infliximab non-responder ulcerative colitis patients.
- Conducted the supporting literature review across IBD immunopathology, anti-TNF resistance, and macrophage biology to define the research question and in vitro study design.
- Gained practical grant-writing experience: building rationale, feasible design, and defined readouts into a fundable proposal.

Laboratory Supervisor, Shahid Beheshti University of Medical Sciences, Tehran, Iran *2017 – 2019*

- Teaching practical techniques in immunology to master students.
- Instructing students on how to operate laboratory instruments.
- Advising and guidance of the master's student. Cooperating with students who do research in the laboratory. Taking exams from students.

M.Sc. Researcher, Shahid Beheshti University of Medical Sciences, Tehran, Iran *2013 – 2016*

- Studied regulatory T cells in esophageal tissue from patients with eosinophilic esophagitis, GERD, and controls; the work was published in Allergologia et Immunopathologia.

PUBLICATIONS

Original articles

- **Mousavinasab F**, Szurek EA, Kuczma MP, Cebula A, Murray JA, Marietta E, Ignatowicz L. Investigating the role of peripheral regulatory T cells in oral tolerance to gluten through the development of a novel mouse model of celiac disease. (Frontiers in Immunology, under review)
- **Mousavinasab F**, Szurek EA, Cebula A, Ignatowicz L, Kuczma MP. Improved protocol for establishing CD4+ hybridomas specific for human class II MHC/peptide complex. Bio-protocol. 2026;16(13):e5737. doi:10.21769/BioProtoc.5737.
- Khatami S, Ebrahimi S, Mahjoub FE, Yousefi A, Hosseini AH, Khoshmirsafa M, **Mousavinasab F**, Mansouri M, Shabani M, Mesdaghi M. Detection of eosinophilic cell-free granules based on expression of CCR3 and MBP markers in colonic biopsy of pediatric patients with suspected and confirmed eosinophilic colitis. Iran J Immunol. 2026;23(2). doi:10.22034/iji.2026.109789.3152.
- Ebrahimi S, Khatami S, Mahjoub FE, Monajemzadeh M, Yousefi A, Hosseini A, Khoshmirsafa M, **Mousavinasab FS**, Mansouri M, Shabani M, Mesdaghi M. Detection of eosinophilic cell-free granules based on expression of CCR3 and MBP markers in esophageal biopsy specimens of patients with suspected and confirmed eosinophilic esophagitis. Pediatr Gastroenterol Hepatol Nutr. 2025;28(1):19-26. PMID: 39839467.
- **Mousavinasab F**, Babaie D, Nilipour Y, Mansouri M, Imanzadeh F, Dara N, Rohani P, Khatami K, Sayyari A, Khoddami M, Kazemiaghdam M, Mesdaghi M. Increased number of regulatory T cells in esophageal tissue of patients with eosinophilic esophagitis in comparison to gastroesophageal reflux disease and control groups. Allergol Immunopathol. 47(5):431-436.
- Abdolahy M, Rasouli Sh, Babaie D, Dara N, Imanzadeh F, Sayyari A, Rouhani P, Khatami K, Kazemiaghdam M, Nilipour Y, Khoddami M, Ghadimi F, **Mousavinasab F**, Mesdaghi M. Increased regulatory T cells in peripheral blood of children with eosinophilic esophagitis. Gastroenterol Hepatol Bed Bench. 2021 Jan 26:25-30.

Review articles and book chapter

- Aghaei-Zarch SM, Nia AHS, Nouri M, **Mousavinasab F**, et al. The impact of particulate matters on apoptosis in various organs: mechanistic and therapeutic perspectives. Biomed Pharmacother. 2023; 165:115054. PMID: 37379642.
- Hoorzad P, **Mousavinasab F**, Tofigh P, et al. Understanding the lncRNA/miRNA-NFκB regulatory network in diabetes mellitus: from function to clinical translation. Diabetes Res Clin Pract. 2023; 202:110804. PMID: 37369279.
- **Mousavinasab F**, Karimi R, Taheri S, et al. Microbiome modulation in inflammatory diseases: progress to microbiome genetic engineering. Cancer Cell Int. 2023;23:271.
- **Mousavinasab F**, Razani E. Harnessing the potential of vesicular stomatitis virus as a novel therapeutic strategy for cancer treatment. Iran J Blood Cancer. 2023;15(2):117-127.
- Salehi-Sardoei A, **Mousavinasab F**, Sayyed RZ, et al. Endophytic fungi: perspectives for microbial engineering (Ch. 11). In: Biostimulants and Protective Biochemical Agents. Elsevier; 2024:167-220. (Book chapter)

PRESENTATIONS

- Digestive Disease Research Day, Emory-Georgia State University (Sep 4, 2026). Investigating the role of peripheral regulatory T cells in oral tolerance to gluten through the development of a novel mouse model of celiac disease. [Dr. Suresh Sitaraman Award, top 3 abstracts]
- AAI Immunology 2026, Boston, MA (Apr 15-19, 2026). Investigating the role of peripheral regulatory T cells in oral tolerance to gluten in celiac disease. [Poster Presentation Award]
- 4th Annual Graduate Conference for Research, Scholarship, and Creative Activity, Atlanta, GA (Feb 11, 2026). Elevated autoreactive $CD4^{+}Foxp3^{-}$ T-cell clones in humanized HLA-DQ8/$Ab^{-/-}$ mice revealed by Nur77-GFP T-cell hybridomas.

- 2025 Southeast Regional Clinical & Translational Science Conference, Pine Mountain, GA (Mar 5-7, 2025). Creating a novel mouse model to investigate the role of peripheral regulatory T cells in oral tolerance to gluten in celiac disease.
- 3rd Annual Graduate Conference for Research, Scholarship, and Creative Activity, Atlanta, GA (Feb 5, 2025). Establishing a novel mouse model to examine how peripheral regulatory T cells contribute to oral tolerance to gluten in celiac disease.
- 2nd Research Day, Institute for Biomedical Sciences, Georgia State University. Developing a novel murine model to define the contribution of peripheral Tregs to gluten-specific oral tolerance in celiac disease.

HONORS & AWARDS

- Dr. Suresh Sitaraman Award, top three abstracts, Digestive Disease Research Day, Emory–Georgia State University (Sep 2026)
- Poster Presentation Award, AAI Immunology 2026, Boston, MA
- Best Oral Presentation, Basic and Advanced Translational Immunology Course, Jaipur, India (Oct 2019)
- FIMSA travel grant, Basic and Advanced Translational Immunology Course, Jaipur, India (2019)
- EFIS travel grant, SEEIS 2018, Yerevan, Armenia
- EAACI junior member travel grant, Helsinki (2017)

TECHNICAL SKILLS

Immunology & cell biology: Multicolor flow cytometry (FlowJo) and adjusting compensation matrix, FACS/MACS cell sorting, primary cell culture, dendritic cell generation, T-cell hybridoma generation and peptide testing.
Molecular biology: DNA/RNA isolation, PCR, real-time qPCR, primer design, cDNA synthesis, gel electrophoresis.
Protein & assays: ELISA, Western blot, SDS-PAGE, BCA, exosome isolation.
Histology & imaging: Tissue processing, H&E staining, immunohistochemistry, immunofluorescence, ImageJ.
In vivo models: Mouse colony management, disease induction, injections, gavage, surgery, organ isolation.
Microbiology: Gram staining of bacteria, Capsule staining of bacteria, Biochemical tests: catalase, oxidase, coagulase, Colony count, Disc diffusion, and well diffusion tests for antibacterial resistance
Quality (QC/QA): ISO 17025 quality control, laboratory documentation and standards, instrument operation and training, analytical result interpretation.
Data & software: GraphPad Prism, SPSS, REST, FlowJo, ImageJ, Adobe Photoshop, EndNote; NCBI/BLAST, genome and protein databases.

MEMBERSHIP AND SERVICE

- Vice President, Translational Biomedical Sciences Student Association, Georgia State University.
- Judge, Georgia State Undergraduate Research Conference (April 2024 and April 2025).
- Peer reviewer, Cellular, Molecular and Biomedical Reports (2023): six manuscripts. https://www.webofscience.com/wos/author/record/47355164
- Member: American Association of Immunologists (AAI); International Union of Immunological Societies (IUIS); European Academy of Allergy and Clinical Immunology (EAACI); Women Speakers, FIMSA.

RESEARCH INTERESTS

- Autoimmunity and tolerance: thymic and peripheral regulatory T cells; tolerance to self and dietary antigens.
- Inflammatory disease: immunoregulation in IBD and GI disorders; macrophage polarization.
- Allergy and cancer immunology: immune profiling in allergy; NK cells and exosomes in hematologic malignancies.

REFERENCES

Leszek Ignatowicz, Ph.D., Distinguished University Professor, Center for Translational Immunology, Georgia State University | lignatowicz@gsu.edu | 404-413-6685

Cynthia Nau Cornelissen, Ph.D., Distinguished University Professor; Director, Center for Translational Immunology; Associate Director, Institute for Biomedical Sciences, Georgia State University | ccornelissen@gsu.edu | 404-413-6680

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